

Financial Inclusion in Europe

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1. Introduction

Financial inclusion refers to the possibility for individuals to access and use financial services, such as an account, digital payments, saving instruments and credit, which can help them to manage everyday transactions and face unexpected financial needs. Even if Europe is generally characterised by high account ownership, this does not mean that all countries or all individuals use financial services in the same way, since differences may remain depending on income, education, gender, age or workforce situation. For this reason, looking only at the average share of people owning an account may give an incomplete image of financial inclusion.

This research studies the topic from two connected perspectives, as it first considers how account ownership varies across the 38 European economies included in the sample, and then examines how demographic and socioeconomic characteristics are related to different financial outcomes within countries. The research question is therefore: How does account ownership vary across European economies, and how are demographic and socioeconomic characteristics associated with different dimensions of financial inclusion within countries?

The analysis uses the updated individual-level release of the Global Findex 2021 Database and focuses on 38,600 respondents from 38 European economies. Weighted descriptive statistics are first used to describe the sample, the diffusion of the main financial indicators and the differences in account ownership between countries, while linear probability models with country fixed effects are used to study the relationship between individual characteristics and account ownership, digital payments, saving and borrowing. The results show that education, income and workforce participation are positively associated with account ownership, digital payment use and saving, although these relationships are not equally strong for all outcomes. In particular, differences across income groups are especially large for saving, while the gender difference in account ownership becomes smaller and is no longer statistically significant after education, income and workforce status are included in the model.

2. Literature

Existing literature generally describes financial inclusion as both access to formal financial services and the possibility to use them, since accounts can support payments, saving, the management of resources and the ability to face unexpected economic needs [1, 2, 3]. Allen et al. [1] analyse financial inclusion using individual-level Global Findex data and show that it is related not only to individual characteristics, but also to the environment in which financial services are provided. In particular, account ownership and use are associated with factors such as the cost of accounts, the proximity of financial institutions, legal protection and political stability, while important differences also exist according to income, gender, education and age. This is particularly relevant for the present analysis, since it shows why individual-level data are useful to study differences that cannot be observed by looking only at aggregate country indicators.

In the European context, Demirgüç-Kunt, Hu and Klapper [2] show that financial inclusion remains quite different across the Europe and Central Asia region, even if account ownership is already very high in most euro-area economies. The authors find that account ownership increased considerably in several developing economies in the region, but the progress was not equal across countries, and gender and income differences

still remained particularly relevant in some cases. The Global Findex 2021 Report [3] confirms this more general pattern, showing that gaps associated with gender, income, age, education and workforce participation remain across world regions, and that the growth in account ownership has not been the same in all economies. These findings suggest that high regional averages can hide both differences between countries and inequalities between individuals living within the same country.

Another important point in the literature is that owning an account does not necessarily mean using it regularly or obtaining benefits from it. Dupas et al. [4] study an experiment in Uganda, Malawi and Chile in which previously unbanked individuals were offered basic accounts without opening or maintenance costs. Although many participants opened the accounts, only a relatively small part used them regularly, and the authors did not find clear average effects on total saving or on other economic outcomes. Their results suggest that simply removing the formal barriers to account ownership may not be sufficient when individuals have very limited resources, when financial products do not correspond to their needs, or when other practical barriers continue to exist. The Global Findex 2021 Report [3] reaches a similar general conclusion, arguing that ownership is an important first step, but that policies also need to support the effective use of accounts for payments, saving and credit.

The distinction between ownership and effective use may also be important when considering gender. Field et al. [5], in a study conducted among rural women in India, compare the simple opening of an individual account with an intervention that also included the direct deposit of wages and training on account use. Women who received both direct deposits and training increased their account use and banking autonomy, and they were more comfortable going to the bank and making transactions independently. Although the institutional and social context is very different from Europe, the study provides useful evidence that formal account ownership does not always imply control over the money in the account or the ability to use it independently.

Building on this literature, the present research report considers account ownership together with digital payments, saving and borrowing, rather than treating financial inclusion as only the possession of an account. It also focuses on demographic and socioeconomic differences within European economies, while recognising that the analysis identifies associations and does not provide causal effects.

3. Data

The Global Findex Database is a data source published by the World Bank [3]. Since 2011, it has gathered granular data on individual behaviour related to saving, borrowing, payment, and financial risk management, at both the global and regional levels.

Originally, the Global Findex 2021 dataset was based on national representative surveys of about 130,000 adults in over 120 economies. The data used in this project is sourced from the updated individual-level release of the Global Findex 2021 Database. It is composed of interviews with individuals aged 15 or older, in which the data are weighted to represent the national adult population and cover 139 economies worldwide, with 143,887 individuals interviewed.

The standardized questionnaire covers access to financial instruments, the use of digital payments, savings, credit, and financial resilience. A part of the survey was conducted in person, however in some cases face-to-face interviews were not possible because of COVID-19-related restrictions on mobility, and telephone surveys were used instead.

The analysis is restricted to 38 European economies. Since the World Bank regional classifications do not correspond exactly to geographic Europe, the countries were selected manually. The complete list of economies included in the sample is reported in Appendix Table A.1. The resulting European sample contains 38,600

respondents. It includes both European Union and non-European Union economies and should therefore not be interpreted as a euro-area sample.

The main outcome is account ownership. Digital payment use, saving, and borrowing are considered as additional dimensions of financial inclusion. The explanatory variables include gender, age, education, income quintile, and workforce status. All financial outcomes are coded as binary variables.

After removing observations with missing information in the variables included in the main regression, the final estimation sample contains 38,302 respondents from all 38 economies.

Account identifies respondents who own an account at a financial institution or through a mobile money provider. Any digital payment indicates whether an individual made or received a digital payment during the previous year. Saved and borrowed identify whether the respondent saved or borrowed money during the same period. The borrowing indicator includes different possible sources of credit and should not be interpreted as an unambiguously positive measure of financial inclusion.

The main explanatory variables are gender, age, education, income quintile, and workforce status. Gender is represented by a dummy variable equal to one for women and zero for men. Age is measured in years. Education is divided into three categories: primary education or less, secondary education, and tertiary education or more. Income is measured using five within-country income quintiles, while workforce status distinguishes between individuals who are in the workforce and those who are out of the workforce.

Survey weights are applied in both the descriptive statistics and the regression analysis. These weights account for the survey sampling structure and help make the results more representative of the adult population within each economy.

4. Methodology

4.1 Descriptive analysis

The descriptive analysis presents the main characteristics of the European sample, including age, gender, education, income, workforce participation, and internet access. It also calculates the weighted shares of account ownership, digital payment use, saving, and borrowing. Finally, account ownership is calculated separately for each economy and presented in a ranked country-level figure. This comparison provides an overview of the differences in basic financial access across the 38 European economies and addresses the first part of the research question. Survey weights are applied throughout the descriptive analysis.

4.2 Model

The specification used is a linear probability model with different financial inclusion indicators as dependent variable, survey weights, country fixed effect and standard errors clustered by country.

A linear probability model was chosen because it provides a simple and transparent structure for binary outcomes and its coefficients can be directly interpretable as probability variations. Age is included in both linear and quadratic form (centred at 45 years before constructing the quadratic term) to allow for a non-linear relationship between age and financial inclusion. Education and income are entered as sets of indicator variables. The reference categories are primary education or less and the poorest income quintile, respectively. Workforce status is represented by a dummy variable, with individuals out of the workforce as the reference group.

Initially, a simpler specification is used with account as dependent variable, and female, age and age squared as independent ones. The main model adds among the regressors education, income quintile, and workforce status. The same specification is estimated for digital payments, savings, and borrowing.

$$Y_{ic} = \alpha_c + \beta_1 Female_{ic} + \beta_2 Age_{ic} + \beta_3 Age_{ic}^2 + \beta_4 Education_{ic} + \beta_5 Income_{ic} + \beta_6 Workforce_{ic} + \varepsilon_{ic} \quad (1)$$

where:

- i denotes the individual;
- c denotes the country;
- Y_{ic} is one of the four outcomes;
- α_c represents the country fixed effects.

Country fixed effects control for characteristics common to individuals from the same economy, such as institutions, level of financial development, infrastructure, general availability of banking services. The coefficients are therefore derived from comparisons between individuals within the same economy.

Standard errors are clustered by country to allow for correlation of errors among individuals belonging to the same economy. The estimates represent conditional associations and should not be interpreted as causal effects.

5. Results

This section presents the main results of the descriptive and regression analyses. It first describes the European sample and the overall diffusion of the four financial inclusion indicators. It then compares account ownership across economies and presents the results of the linear probability models.

5.1 Descriptive results

The European sample includes 38,600 respondents from 38 economies. The weighted average age is approximately 46.6 years, and women represent 51.5 percent of the sample. Around 60.1 percent of respondents are in the workforce, while the majority of the sample has secondary education, which accounts for 59.9 percent.

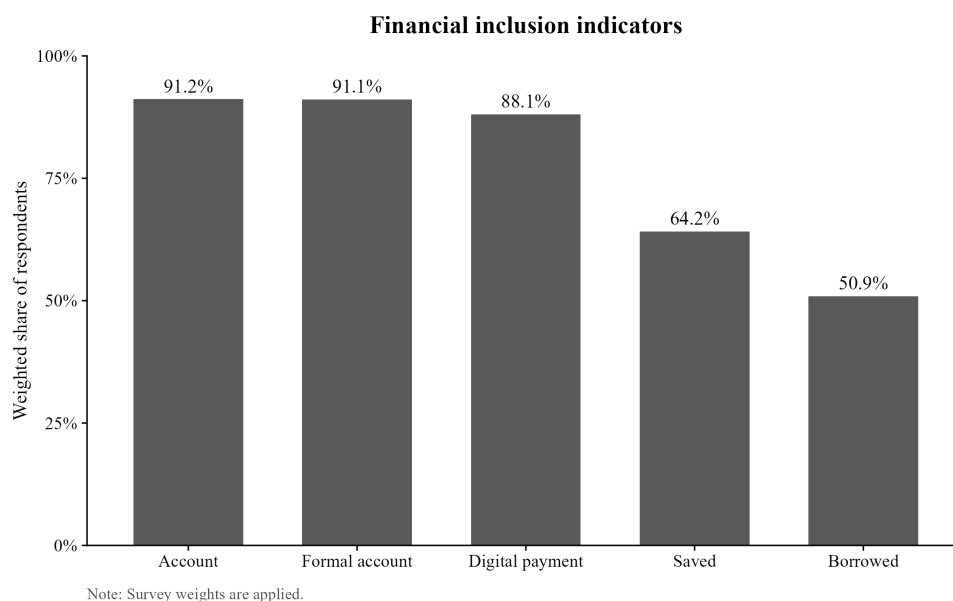


Figure 1: Financial indicators

Note: Formal account ownership is reported because it was included in the initial descriptive analysis. However, since it is almost identical to overall account ownership in the European sample, it is not considered as a separate outcome in the regression analysis.

The descriptive results show that account ownership and digital payment use are widespread in the European sample. Approximately 91.2 percent of respondents report owning an account, while 88.1 percent report having made or received a digital payment. Saving and borrowing are less common: around 64.2 percent of respondents report having saved money, while 50.9 percent report having borrowed during the previous year.

There is therefore a clear difference between basic access to a financial account and other financial behaviours. Although most respondents own an account and use digital payments, the shares reporting saving or borrowing are considerably lower.

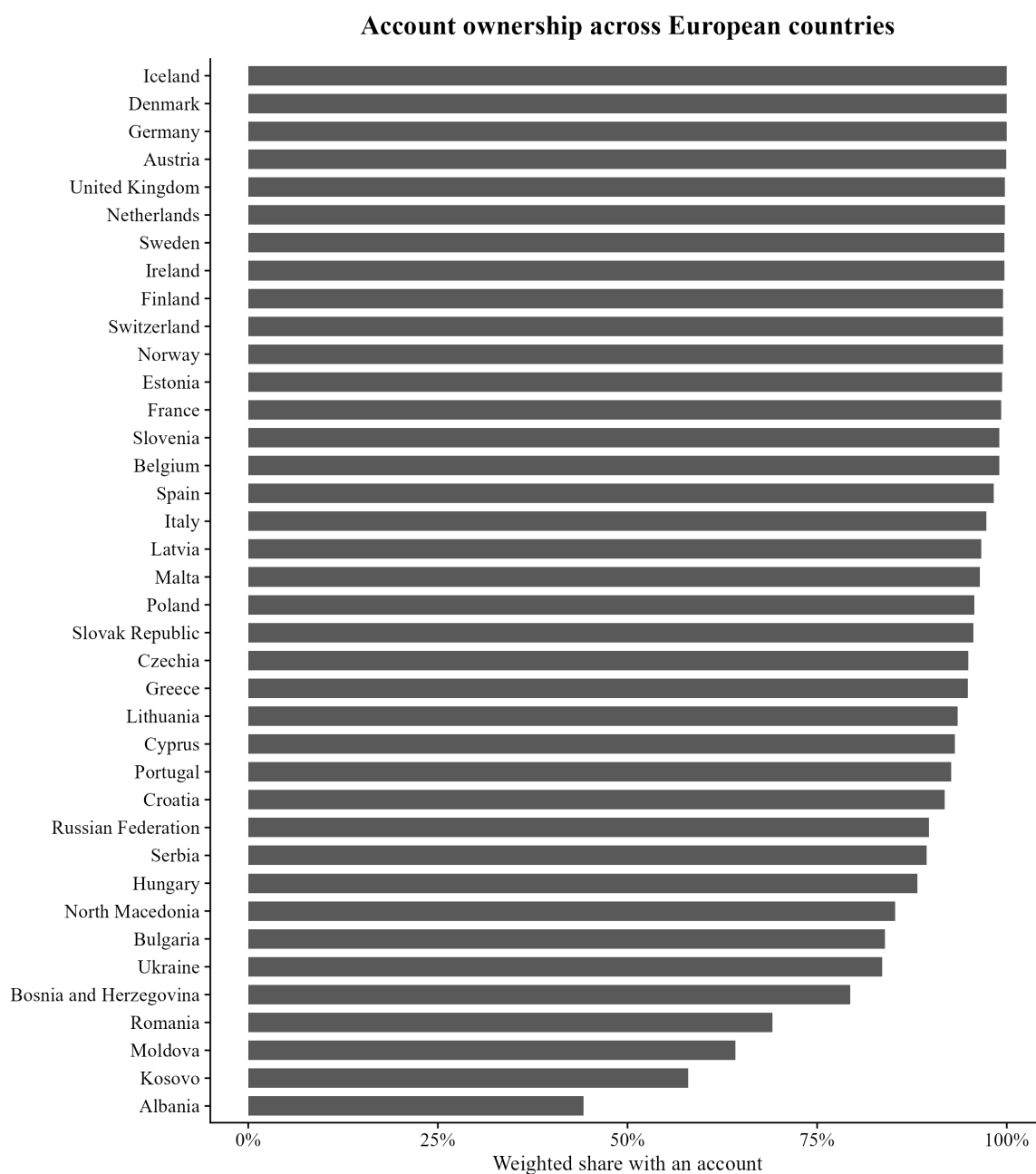


Figure 2: Account ownership across European economies

Figure 2 presents the weighted share of adults owning an account in each of the 38 European economies. Account ownership is close to universal in several countries, but the figure also shows substantial differences across the sample. This means that the generally high level of account ownership observed in Europe does not apply equally to all economies. The highest levels are observed in northern European countries, while several eastern and southeastern European countries show lower rates in the sample.

5.2 Main regression results

The descriptive analysis shows differences both across economies and across financial inclusion indicators. The regression analysis examines whether individual demographic and socioeconomic characteristics are associated with these outcomes after accounting for average differences across countries.

Table 1: Account ownership: main regression results

Dependent variable: Account ownership	(1)	(2)
	Demographic model	Main model
Female	−0.020* (0.009)	−0.009 (0.007)
Age (centered)	0.0001 (0.0003)	0.0004 (0.0004)
Age squared	$-8.14 \times 10^{-5}***$ (1.55×10^{-5})	$-3.77 \times 10^{-5}**$ (1.20×10^{-5})
Secondary education		0.079*** (0.016)
Tertiary education		0.096*** (0.018)
Income quintile 2		0.029*** (0.008)
Income quintile 3		0.047*** (0.009)
Income quintile 4		0.054*** (0.011)
Richest income quintile		0.064*** (0.013)
In the workforce		0.051*** (0.013)
Country fixed effects	Yes	Yes
Observations	38,302	38,302
R^2	0.20683	0.23820

Notes: Weighted linear probability models. Standard errors clustered by economy are reported in parentheses. The reference categories are male, primary education or less, the poorest income quintile, and out of the workforce. Both specifications include country fixed effects. Age is centered at 45 years.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 1 reports the results for account ownership. The first specification includes gender, age, age squared, and country fixed effects. The second specification adds education, income quintile, and workforce status and represents the main model.

In the main specification, individuals with secondary education have an estimated probability of owning an account approximately 7.9 percentage points higher than individuals with primary education or less and the result is statistically significant at $p < 0.001$. Individuals with a tertiary education are approximately 9.6 percentage points, statistically significant, more likely to have an account than those with a primary education or less, holding all other characteristics constant.

Income is also positively associated with account ownership. Relative to individuals in the poorest income quintile, those in progressively richer quintiles have a higher conditional probability of owning an account, specifically 6.4 percentage points compared to the poorest quintile.

Finally, women have an estimated probability of owning an account that is approximately 2 percentage points lower than that of men. After education, income, and workforce status are added, the estimated difference decreases to approximately 0.9 percentage points and is no longer statistically significant.

5.3 Alternative outcomes

Table 2 reports the results obtained by applying the main specification to digital payment use, saving, and borrowing. Overall, the results show that the relationships between individual characteristics and financial inclusion differ depending on the outcome considered.

The strongest socioeconomic differences are observed for saving. Education and income are both positively associated with the probability of saving, with the income gradient being particularly large. Individuals in the richest income quintile have an estimated probability of saving approximately 21.3 percentage points higher than those in the poorest quintile. Tertiary education is associated with an increase of around 14.6 percentage points. Women are approximately 1.9 percentage points less likely to report saving than men, while workforce participation is positively associated with saving. These results indicate that the ability to save differs more strongly across income groups than the other outcomes considered.

Borrowing shows different results as education and income are not statistically significant at the 5 percent level, and there is no clear pattern across the different categories. Women are approximately 3.1 percentage points less likely to report borrowing, while individuals in the workforce have an estimated probability around 6.2 percentage points higher than those outside the workforce.

6. Discussion

The results suggest that a high level of account ownership does not necessarily mean that financial services are used in the same way by all individuals. Account ownership and digital payment use are widespread in the European sample, but saving is less common and shows larger differences across socioeconomic groups. This indicates that financial inclusion should not be evaluated only through access to an account, but also through the way individuals use financial services.

Education and income appear to be particularly relevant. Individuals with higher education and those belonging to richer income groups are more likely to own an account, use digital payments, and save. Education may be associated with greater familiarity with financial and digital services, as well as with different employment opportunities. Income is especially important for saving, since individuals with fewer economic resources may have a more limited ability to set money aside. However, these results describe associations and do not show that education or income directly cause higher financial inclusion.

Table 2: Financial inclusion outcomes

Dependent variable	Account ownership	Digital payments	Saving	Borrowing
Model	(1)	(2)	(3)	(4)
Female	−0.009 (0.007)	−0.008 (0.007)	−0.019** (0.007)	−0.031*** (0.008)
Age (centered)	0.0004 (0.0004)	−0.0001 (0.0004)	−0.003*** (0.0003)	−0.002*** (0.0003)
Age squared	-3.77×10^{-5} ** (1.20×10^{-5})	-4.17×10^{-5} *** (1.16×10^{-5})	5.27×10^{-5} * (1.95×10^{-5})	−0.0001*** (1.44×10^{-5})
Secondary education	0.079*** (0.016)	0.095*** (0.016)	0.086*** (0.014)	4.51×10^{-5} (0.018)
Tertiary education	0.096*** (0.018)	0.122*** (0.020)	0.146*** (0.017)	0.038 (0.021)
Income quintile 2	0.029*** (0.008)	0.045*** (0.009)	0.071*** (0.010)	−0.012 (0.012)
Income quintile 3	0.047*** (0.009)	0.073*** (0.010)	0.130*** (0.013)	−0.007 (0.012)
Income quintile 4	0.054*** (0.011)	0.077*** (0.011)	0.172*** (0.016)	0.007 (0.013)
Richest income quintile	0.064*** (0.013)	0.092*** (0.014)	0.213*** (0.019)	0.025 (0.015)
In the workforce	0.051*** (0.013)	0.058*** (0.015)	0.054*** (0.010)	0.062*** (0.011)
Country fixed effects	Yes	Yes	Yes	Yes
Observations	38,302	38,302	38,302	38,302
R ²	0.23820	0.27049	0.20424	0.08494

Notes: Weighted linear probability models. Standard errors clustered by economy are reported in parentheses. The reference categories are male, primary education or less, the poorest income quintile, and out of the workforce. All specifications include country fixed effects. Age is centered at 45 years.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; · $p < 0.10$.

The results for gender are more mixed. After education, income, and workforce status are included, there is no statistically significant difference between women and men in account ownership or digital payment use. Differences remain, however, for saving and borrowing, with women being less likely to report both activities. This suggests that similar levels of access do not necessarily imply similar financial behaviour or an equal availability of financial resources. Other factors not observed in the dataset, such as financial control within the household or differences in employment and income stability, may also be relevant.

Education and income are not clearly associated with the probability of borrowing, and this may be because borrowing is not always a positive indicator of financial inclusion, as it can reflect access to credit, but also financial need or difficulty in covering expenses. Moreover, the variable includes borrowing from different sources, which may be formal or informal, so it should be interpreted differently from account ownership, digital payments, and saving.

From a policy perspective, the findings suggest that expanding account ownership alone may not be sufficient. Policies could also focus on the effective use of financial services, including financial and digital skills. Individuals with lower education or income may face barriers even when they formally own an account. Therefore, financial inclusion policies should distinguish between access to financial services and the ability to use them in a way that responds to individual needs.

7. Limitations

This analysis has several limitations. First, the data are cross-sectional and observational, meaning that they capture individuals at one point in time and do not allow for causal conclusions, so the estimated coefficients can only show associations between individual characteristics and financial inclusion outcomes.

Second, the models may omit relevant individual characteristics that are not included in the dataset or in the final specification. These may include financial literacy, urban or rural location, family background, type of occupation, trust in financial institutions, and the quality of digital infrastructure. If these characteristics are related to both the explanatory variables and the financial outcomes, they may partly bias the estimated relationships.

The analysis also relies on self-reported information. Respondents may not remember past financial activities precisely, may interpret survey questions differently, or may choose not to report some behaviours accurately. In addition, the indicators provide only a limited picture of financial inclusion. Account ownership does not measure how frequently an account is used, saving does not indicate the amount saved, and borrowing includes different motivations and both formal and informal sources. These outcomes therefore do not directly measure financial well-being.

Finally, the European sample includes economies with different income levels, financial systems, institutions, and infrastructures. Country fixed effects absorb average differences across economies, but they do not explain which national characteristics are responsible for the observed cross-country variation. The data were also collected during the COVID-19 period, which may have accelerated the use of digital payments and temporarily changed saving and borrowing behaviours.

8. Conclusion

This research report examines financial inclusion across 38 European economies using individual-level data from the Global Findex 2021 Database, with the aim of understanding both how account ownership differs across countries and how demographic and socioeconomic characteristics are related to different dimensions of financial inclusion within them. Although account ownership is widespread in most of the economies considered, the descriptive analysis still reveals relevant cross-country differences, while the regression results show that education, income, and workforce status are positively associated not only with account ownership, but also with digital payment use and saving, even after accounting for average differences across countries.

Among the main results, the strongest socioeconomic differences emerge for saving, especially across income groups, while education is consistently associated with a higher probability of using financial services. The gender gap in account ownership becomes smaller and statistically insignificant once education, income, and workforce status are included in the model, whereas differences remain for saving and borrowing. Borrowing, however, follows a different pattern from the other outcomes, since education and income do not show the same clear relationship observed for account ownership, digital payments, and saving, which may reflect the fact that borrowing can be linked both to access to credit and to financial need.

Overall, these findings suggest that financial inclusion should not be evaluated only through account ownership, since having access to an account does not necessarily imply that individuals use financial services in the same way or have the same ability to save. Policies aimed at promoting financial inclusion should therefore consider not only formal access, but also the effective use of financial services, with particular attention to individuals with lower income or education. At the same time, the results show relationships between the variables, but not causal effects, and future research could extend the analysis by comparing different waves of the Global Findex

Database or by including country-level characteristics to better understand changes over time and the factors behind the differences observed across economies.

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A. Sample Composition

Table A.1: European economies included in the analysis

No.	Economy	No.	Economy
1	Albania	20	Latvia
2	Austria	21	Lithuania
3	Belgium	22	Malta
4	Bosnia and Herzegovina	23	Moldova
5	Bulgaria	24	Netherlands
6	Croatia	25	North Macedonia
7	Cyprus	26	Norway
8	Czechia	27	Poland
9	Denmark	28	Portugal
10	Estonia	29	Romania
11	Finland	30	Russian Federation
12	France	31	Serbia
13	Germany	32	Slovak Republic
14	Greece	33	Slovenia
15	Hungary	34	Spain
16	Iceland	35	Sweden
17	Ireland	36	Switzerland
18	Italy	37	Ukraine
19	Kosovo	38	United Kingdom

Notes: The sample includes European Union and non-European Union economies. The economies were selected manually because the World Bank regional classifications do not correspond exactly to geographic Europe.

B. Additional Figures

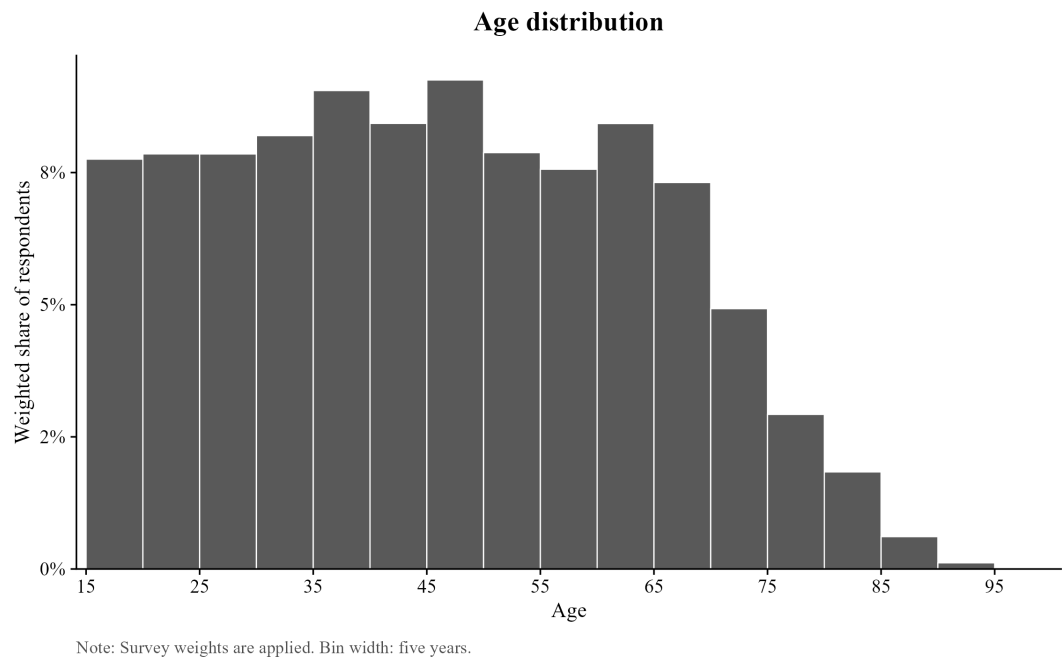


Figure B.1: Age distribution in the European sample

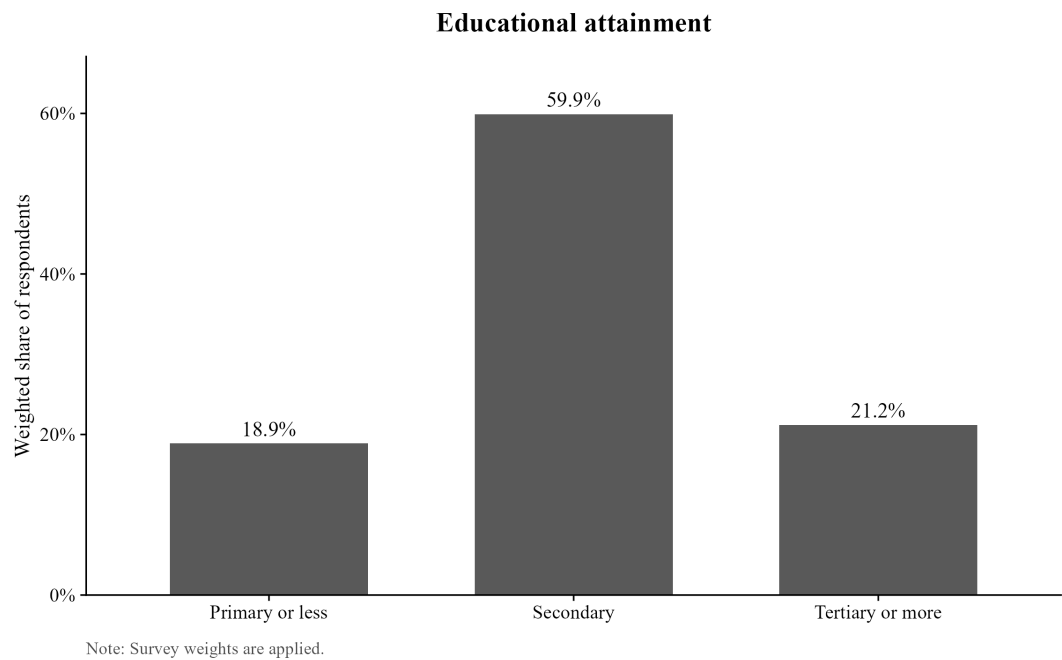


Figure B.2: Education distribution in the European sample

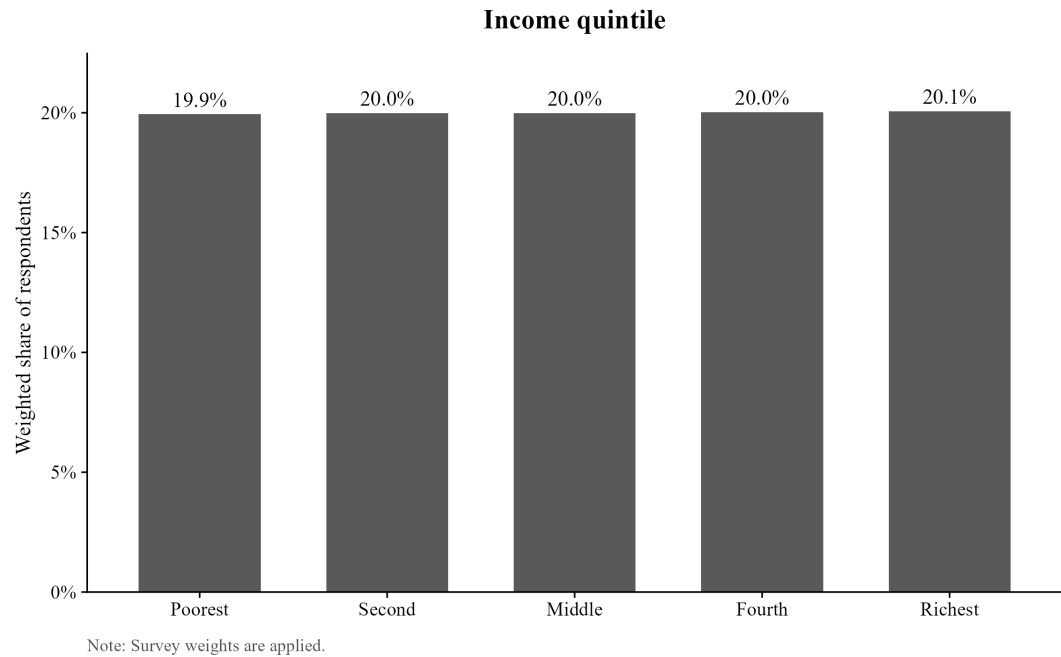


Figure B.3: Income distribution in the European sample

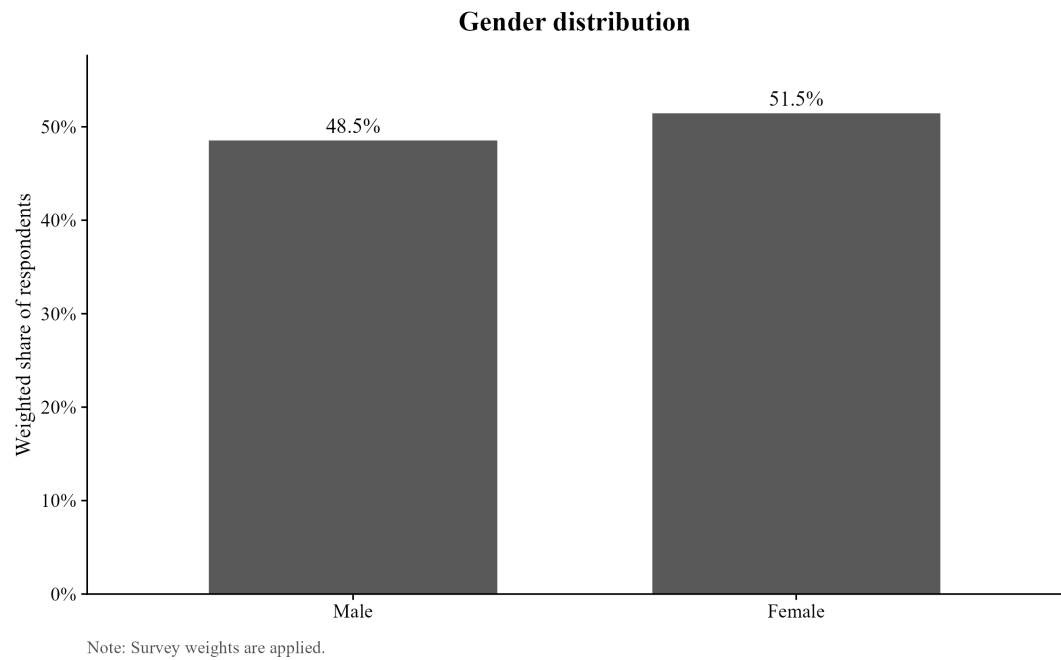


Figure B.4: Gender distribution in the European sample

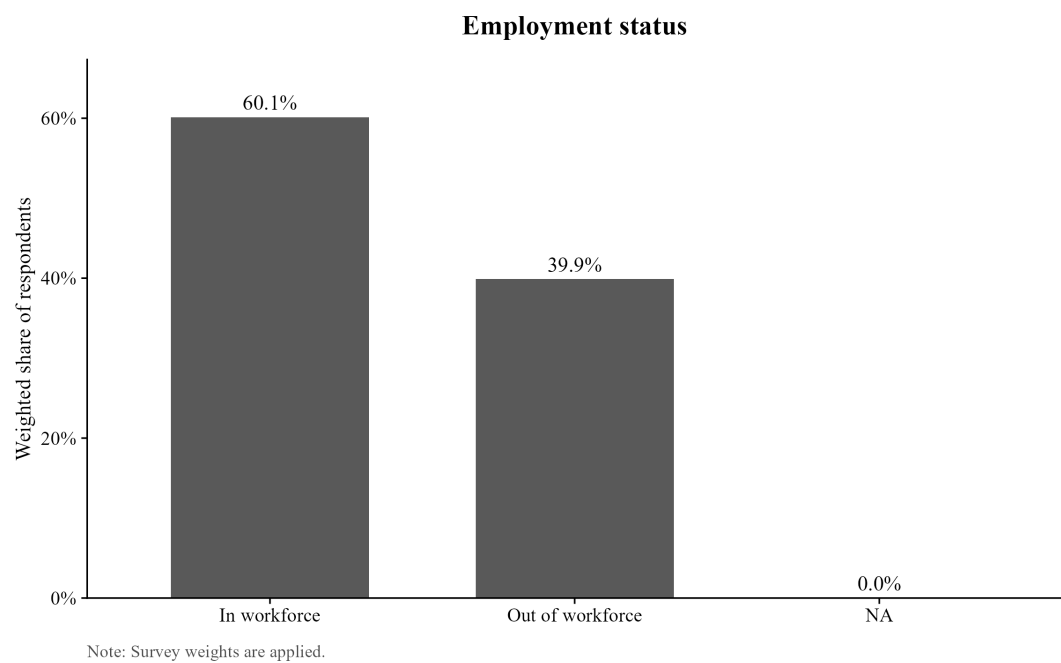


Figure B.5: Employment status in the European sample